

Question 19

Correct

Mark 1.00 out of 1.00

Flag question

In empirical risk minimization the predictor is:

Select one:

- ☐ a. None of the above.
- ☐ b. A probabilistic function.
- ☒ c. A deterministic function.



Question 20

Correct

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Flag question

The regression matrix A of the data (x_i, y_i) , $i = 1, \dots, N$, $x_i, y_i \in \mathbb{R}$, for a polynomial model of degree $k - 1$, has elements:

Select one:

- ☐ a. $a_{i,j} = x_i^{j+1}$, $i = 1, \dots, N$, $j = 1, \dots, k$.
- ☒ b. $a_{i,j} = x_i^{j-1}$, $i = 1, \dots, N$, $j = 1, \dots, k$.
- ☐ c. $a_{i,j} = x_i^j$, $i = 1, \dots, N$, $j = 1, \dots, k$.



Question 21

Question 17

Correct

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Flag question

If X, Y are random variables with values in $\mathbb{R}^D$, then:

Select one:

- ☒ a. $Cov(X, Y)$ is a matrix $D \times D$.
- ☐ b. $Cov(X, Y)$ is a scalar.
- ☐ c. $Cov(X, Y)$ is a vector $D \times 1$.



Question 18

Correct

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Flag question

Given two random variables X and Y , Bayes Theorem implies that $p(y|x) = \frac{p(x)p(y|x)}{p(x)}$ where:

Select one:

- ☐ a. $p(x|y)$ is called posterior distribution on y .
- ☐ b. $p(x|y)$ is called prior distribution on x .
- ☒ c. $p(x|y)$ is called likelihood on y .



Question 19

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Question 20

Correct

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 Flag question

The regression matrix A of the data (x_i, y_i) , $i = 1, \dots, N$, $x_i, y_i \in \mathbb{R}$, for a polynomial model of degree $k - 1$, has elements

Select one:

- ☐ a. $a_{i,j} = x_i^{j+1}$, $i = 1, \dots, N$, $j = 1, \dots, k$ ☐
- ☒ b. $a_{i,j} = x_i^{j-1}$, $i = 1, \dots, N$, $j = 1, \dots, k$ ☐
- ☐ c. $a_{i,j} = x_i^j$, $i = 1, \dots, N$, $j = 1, \dots, k$ ☐
- 

Question 21

Correct

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 Flag question

If U is an $n \times n$ orthogonal matrix, $x \in \mathbb{R}^n$, then:

Select one:

- ☒ a. None of the above. ☐
- ☐ b. $\|Ux\| > \|x\|$ ☐
- ☐ c. $\|Ux\| < \|x\|$ ☐
- 

Question 15

Correct

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Flag question

For a random variable $X: \Omega \rightarrow \mathcal{T}$, its variance is defined as:

Select one:

- ☒ a. $\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$
- ☐ b. $\text{Var}(X) = \mathbb{E}[X - \mathbb{E}[X]]$
- ☐ c. $\text{Var}(X) = \mathbb{E}[X^2 - \mathbb{E}[X]^2]$



Question 16

Correct

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Flag question

Given a discrete random variable $X: \Omega \rightarrow \mathcal{T}$, with $\mathcal{T} = \{1, 2, 3\}$ and $f_X = \{\frac{1}{6}, \frac{1}{3}, \frac{1}{2}\}$ its PMF, then:

Select one:

- ☒ a. $\mathbb{E}[X] = 2$
- ☐ b. $\mathbb{E}[X] = \frac{11}{6}$
- ☐ c. $\mathbb{E}[X] = 6$

$$\frac{1}{6} + \frac{2}{3} + \frac{3}{2} = \frac{1+4+9}{6}$$



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Question 7

Correct

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Flag question

If $A = U\Sigma V^T$ is the SVD decomposition of an $m \times n$ matrix A , then:

Select one:

- ☐ a. None of the above
- ☒ b. The rows of V^T are eigenvectors of $A^T A$.
- ☐ c. The columns of U are eigenvectors of $A^T A$.

$$A^T A = (U \Sigma V^T)^T (U \Sigma V^T) = V \Sigma^T U^T U \Sigma V^T = V \Sigma^2 V^T$$



Question 8

Correct

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Flag question

The solution of $\min_x \|Ax - b\|_2^2$ where A is an $m \times n$ matrix, $m \geq n$, $\text{rank}(A) = n$, can be computed as:

Select one:

- ☐ a. $AA^T x = A^T b$.
- ☒ b. $A^T A x = A^T b$.
- ☐ c. $A^T A x = b$.



Question 9

Correct

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Flag question

If $f: \mathbb{R}^3 \rightarrow \mathbb{R}$, $f(x_1, x_2, x_3) = \sin x_1 - \sin x_2 \cos x_3 + x_3^2$, then $\nabla f(0, \pi, \pi)$ equals to:

Select one:

Question 13

Incorrect

Mark 0.00 out of 1.00

 Flag question

If $f: \mathbb{R}^n \rightarrow \mathbb{R}$, $f \in C^1(\mathbb{R}^n)$, then the Gradient Descent method $x_{k+1} = x_k - \alpha_k \nabla f(x_k)$ converges to a stationary point of f if:

Select one:

- ☐ a. $\alpha_k \rightarrow 0$ as $k \rightarrow \infty$.
- ☐ b. $\alpha_k > 0 \forall k \in \mathbb{N}$.
- ☒ c. $\alpha_k > 0$ is chosen with a backtracking procedure.



Question 14

Incorrect

Mark 0.00 out of 1.00

 Flag question

If $\mathcal{A}$ is the event space and $\mathcal{T}$ is a subset of $\mathbb{R}$, then:

Select one:

- ☐ a. $\forall A \in \mathcal{A}$, $X(A)$ is the probability that X lies in A .
- ☒ b. None of the above.
- ☐ c. $\forall A \in \mathcal{A}$, $X(A)$ is the event " X lies in A ".



Question 15

Correct

Mark 1.00 out of 1.00

For a random variable $X: \Omega \rightarrow \mathcal{T}$, its variance is defined as:

Select one:

Question 5

Correct

Mark 1.00 out of 1.00

Flag question

If V is a vector space, $U \subseteq V$ is a subspace, and $\Pi_U : V \rightarrow U$ is a projection. Then:

Select one:

- ☐ a. $\Pi_U(x) \notin U, \forall x \in V.$
- ☒ b. None of the above.
- ☐ c. $\Pi_U(x) = 0, \forall x \in V.$



Question 6

Incorrect

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Flag question

If

$$A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

then:

Select one:

- ☒ a. $K_2(A) = 4$
- ☐ b. $K_2(A) = 2$
- ☐ c. $K_2(A) = \frac{1}{2}$



Question 9

Correct

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Flag question

If $f: \mathbb{R}^3 \rightarrow \mathbb{R}$, $f(x_1, x_2, x_3) = \sin x_1 - \sin x_2 \cos x_3 + x_3^2$, then $\nabla f(0, \pi, \pi)$ equals to:

Select one:

- ☒ a. None of the above.
☐ b. $(-1, 0, \pi)$.
☐ c. $(0, 0, 0)$.

$$\begin{aligned}\nabla f &= \cos x_1 \rightarrow 1 \\ &\quad - \cos x_2 \cos x_3 \rightarrow -1 \\ &\quad + \sin x_2 \sin x_3 + 2x_3 \rightarrow 2\pi\end{aligned}$$



Question 10

Correct

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If $f: \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = x_1 x_2$, $g: \mathbb{R} \rightarrow \mathbb{R}^2$, $g(t) = (t, t^2)$ then, if $h(t) = f(g(t))$,

Select one:

- ☐ a. $h'(t) = 3t^2 + 1$.
☐ b. $h'(t) = 3t^2 - 1$.
☒ c. None of the above.

$$\begin{bmatrix} t^2 & t \end{bmatrix} \begin{bmatrix} 1 \\ 2t \end{bmatrix} = t^2 + 2t^2$$



Question 11

Correct

Mark 1.00 out of 1.00

If $f: \mathbb{R}^n \rightarrow \mathbb{R}$, Which of the following statements is False?

Select one:

☐ a. The best approximation for f at x is $\nabla f(x) \cdot (x - x_0)$.

Question 3

Incorrect

Mark 0.00 out of 1.00

Flag question

If $A \in \mathbb{R}^{n \times n}$ is symmetric and positive definite, then:

Select one:

- ☒ a. None of the above.
- ☐ b. $A_{i,j} \geq 0$ for all i, j .
- ☐ c. All the eigenvalues λ_i of A are ≥ 0 .



Question 4

Incorrect

Mark 0.00 out of 1.00

Flag question

If U is an $n \times n$ orthogonal matrix, then:

Select one:

- ☐ a. The rows of U are orthogonal vectors.
- ☒ b. None of the above.
- ☐ c. The rows of U are orthonormal vectors.



Question 5

Correct

Mark 1.00 out of 1.00

If V is a vector space, $U \subseteq V$ is a subspace, and $\Pi_U : V \rightarrow U$ is a projection. Then:

Select one:

Question 1

Incorrect

Mark 0.00 out of 1.00

Flag question

If $x = 0.7 \cdot 10^{-4}$ and $y = 1.7892$, which is the value of $z = x + y$ when represented in $\mathcal{F}(10, 5, -5, 5)$?

Select one:

☐ a. None of the above.☒ b. $0.17892 \cdot 10^1$ ☐ c. $0.17899 \cdot 10^1$

$$x = 0,00007$$

$$y = 1,7892$$

$$z = 1,98927$$



Question 2

Incorrect

Mark 0.00 out of 1.00

Flag question

The 1-norm of a vector x of size n is defined as:

Select one:

☒ a. $\|x\|_1 = \sum_{i=1}^n |x_i|$ ☐ b. None of the above.☐ c. $\|x\|_1 = \max_i |x_i|$ 

Question 3

Incorrect

If $A \in \mathbb{R}^{n \times n}$ is symmetric and positive definite, then:

Question 11

Correct

Mark 1.00 out of 1.00

Flag question

If $f: \mathbb{R}^n \rightarrow \mathbb{R}$. Which of the following statements is False?

Select one:

- ☐ a. x^* local minimum for $f \implies \nabla f(x^*) = 0$.
☐ b. $\nabla f(x^*) = 0 \implies x^*$ stationary point for f .
☒ c. $\nabla f(x^*) = 0 \implies x^*$ local minimum for f .



Question 12

Correct

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Flag question

Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = 9x_1x_2^2 - x_1$, $\nabla f(x_1, x_2) = (9x_2^2 - 1, 18x_1x_2)$ then which of the following is a stationary point for f ?

Select one:

- ☐ a. $(0, 0)$.
☒ b. None of the above.
☐ c. $(0, 3)$.

$$9x_2^2 = 1 \quad x_2 = \pm 1/3$$

$$18x_1x_2 = 0 \quad x_1 = 0$$



Question 13

Incorrect

Mark 0.00 out of 1.00

If $f: \mathbb{R}^n \rightarrow \mathbb{R}$, $f \in C^1(\mathbb{R}^n)$, then the Gradient Descent method $x_{k+1} = x_k - \alpha_k \nabla f(x_k)$ converges to a stationary point of f if.

Select one:

- ☐ a. $\alpha_k \rightarrow 0$ and $\sum \alpha_k = \infty$.

Domanda 11

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Contrassegna
domanda

If $f: \mathbb{R}^n \rightarrow \mathbb{R}$, then:

Scegli un'alternativa:

- ☐ a. x^* local minimum for $f \iff \nabla f(x^*) = 0$.
☐ b. $\nabla f(x^*) = 0 \implies x^*$ local minimum for f .
☒ c. x^* local minimum for $f \implies \nabla f(x^*) = 0$.



Domanda 12

Risposta
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Contrassegna
domanda

Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = 9x_1x_2^2 - x_1$,
 $\nabla f(x_1, x_2) = (9x_2^2 - 1, 18x_1x_2)$ then which of the following is a
stationary point for f ?

Scegli un'alternativa:

- ☒ a. None of the above.
☐ b. $(0, 0)$.
☐ c. $(0, 3)$.



Domanda 13

Risposta
corretta
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Contrassegna
domanda

If $f: \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = x_1e^{x_2}$, then if the initial guess for a gradient
descent iteration is $x^{(0)} = (1, 1)^T$ and $\alpha = \frac{1}{2}$, then:

Scegli un'alternativa:

- ☒ a. $x^{(1)} = (1 - \frac{e}{2}, 1 - \frac{e}{2})^T$.
☐ b. $x^{(1)} = (1 + \frac{e}{2}, 1 + \frac{e}{2})^T$.
☐ c. $x^{(1)} = (\frac{1}{2} - \frac{e}{2}, \frac{1}{2} - \frac{e}{2})^T$.

$$x_1 = (1, 1) - \frac{1}{2} \nabla f(1, 1)$$

=



Domanda 14

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Contrassegna
domanda

Given two random variables X and Y , then the posterior probability of X
given Y is defined as:

Scegli un'alternativa:

- ☐ a. $P(X = x, Y = y)$.
☐ b. $P(Y = y|X)$.
☒ c. $P(X = x|Y)$.



Domanda 15

Risposta
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Contrassegna
domanda

For a random variable $X: \Omega \rightarrow \mathcal{T}$ with $\mathbb{E}[X] = 0$, it holds:

Scegli un'alternativa:

- ☒ a. $\text{Var}(X) = \mathbb{E}[X^2]$.
☐ b. $\text{Var}(X) = \mathbb{E}[X]$.
☐ c. $\text{Var}(X) = 0$.

$$\text{Var}(x) = \mathbb{E}[x^2] - \mathbb{E}[x]^2$$



Domanda 16

Risposta

Given a discrete random variable $X: \Omega \rightarrow \mathcal{T}$, with $\mathcal{T} = \{0, 1\}$, and
 $f_x = \{ \frac{2}{3}, \frac{1}{3} \}$, then:



Domanda 6

Risposta
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Contrassegna
domanda

If

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

then:

Scegli un'alternativa:

- ☒ a. $K_2(A) = \frac{1}{2}$.
- ☐ b. $K_2(A) = 4$.
- ☒ c. $K_2(A) = 1$.



Domanda 7

Risposta errata

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Contrassegna
domandaIf $A \in \mathbb{R}^{n \times n}$, then:

Scegli un'alternativa:

- ☐ a. It depends on A .
- ☒ b. $A^T A$ is symmetric but not necessarily positive definite.
- ☐ c. $A^T A$ is always symmetric and positive definite.



Domanda 8

Risposta errata

Punteggio
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Contrassegna
domandaA solution of $\min_x \|Ax - b\|_2^2$ where A is an $m \times n$ matrix, $m \geq n$, $\text{rank}(A) = k < n$, can be computed as:

Scegli un'alternativa:

- ☐ a. $A^+ x = b$.
- ☐ b. $AA^T x = A^T b$.
- ☒ c. $x = A^+ b$.



Domanda 9

Risposta
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Contrassegna
domanda

$$\begin{bmatrix} 2 \cos t + \sin t & \cos t \end{bmatrix} \begin{bmatrix} \sin t \\ \cos t \end{bmatrix} = -2 \sin \cos - \sin^2 + \cos^2$$

If $f: \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = x_1^2 + x_1 x_2$, $g: \mathbb{R} \rightarrow \mathbb{R}^2$, $g(t) = (\cos(t), \sin(t))$, then, if $h(t) = f(g(t))$:Scegli un'alternativa: $\begin{bmatrix} 2x_1 + x_2 & x_1 \end{bmatrix} \begin{bmatrix} -\sin t \\ \cos t \end{bmatrix}$

- ☐ a. $h'(t) = \cos(2t) - 1$.
- ☐ b. $h'(t) = 1 - \sin(2t)$.
- ☒ c. $h'(t) = \cos(2t) - \sin(2t)$.



Domanda 10

Risposta
correttaPunteggio
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Contrassegna
domandaIf $f: \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = x_1 + 2x_2^2$, $g: \mathbb{R} \rightarrow \mathbb{R}^2$, $g(t) = (t, t^2)$ then, if $h(t) = f(g(t))$,

$$\begin{bmatrix} 1 & 4x_2 \end{bmatrix} \begin{bmatrix} 1 \\ 2t \end{bmatrix}$$

Scegli un'alternativa:

- ☒ a. $h'(t) = 8t^3 + 1$.
- ☐ b. $h'(t) = t^3 + t$.
- ☐ c. $h'(t) = 3t^3 + 1$.

$$1 + 8t^3$$



Domanda 11

Domanda 16

Risposta
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Contrassegna
domanda

Given a discrete random variable $X: \Omega \rightarrow \mathcal{T}$, with $\mathcal{T} = \{0, 1\}$, and $f_X = \{\frac{2}{3}, \frac{1}{3}\}$, then:

Scegli un'alternativa:

- ☐ a. $\mathbb{E}[X] = 1$
- ☐ b. $\mathbb{E}[X] = \frac{2}{3}$
- ☒ c. $\mathbb{E}[X] = \frac{1}{3}$



Domanda 17

Risposta
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Contrassegna
domanda

If X, Y are multivariate random variables with states $x, y \in \mathbb{R}^D$, then:

Scegli un'alternativa:

- ☐ a. $Cov(X, Y) = \mathbb{E}[X, Y^T] - \mathbb{E}[X]$
- ☐ b. $Cov(X, Y) = \mathbb{E}[X, Y] + \mathbb{E}[X]\mathbb{E}[Y]$
- ☒ c. $Cov(X, Y) = Cov(Y, X)$



Domanda 18

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Contrassegna
domanda

Given two random variables X and Y , Bayes Theorem implies that $p(y|x) = \frac{p(x|y)p(y)}{p(x)}$ where:

Scegli un'alternativa:

- ☒ a. $p(x|y)$ is called likelihood on y .
- ☐ b. $p(x|y)$ is called prior distribution on x .
- ☐ c. $p(x|y)$ is called posterior distribution on y .



Domanda 19

Risposta
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Contrassegna
domanda

Given two random variables X and Y such that $p(x) = ce^{-|x|}$ and $p(y|x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(y-ax)^2}$, then the MLE reads:

Scegli un'alternativa:

- ☐ a. $x^* = \arg \min_x \frac{1}{2}(y - ax)^2 + \frac{1}{2}x^2$
- ☐ b. $x^* = \arg \min_x \frac{1}{2}(y - ax)^2 + |x|$
- ☒ c. $x^* = \arg \min_x \frac{1}{2}(y - ax)^2$



Domanda 20

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Suppose a set of data (x_i, y_i) , $i = 1, \dots, N$, $y_i = f(x_i) + \epsilon_i$, where $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$. In linear regression, the likelihood function is:

Scegli un'alternativa:

- ☐ a. $p(y|x, \theta) = \prod_{i=1}^N \mathcal{N}(x|x_i^T \theta, \sigma^2)$
- ☒ b. None of the above.
- ☐ c. $p(y|x, \theta) = \prod_{i=1}^N \mathcal{N}(y|x_i^T \theta x_i, \sigma^2)$



Domanda 21

Risposta errata

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A solution of $\min_x \|Ax - b\|_2^2$ where A is an $m \times n$ matrix, $m \geq n$, $\text{rank}(A) = k < n$, can be computed as:

Scegli un'alternativa:

- ☐ a. $x = A^+ b$



Domanda 1

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Contrassegna
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If $x = 3.89167$ and $y = 0.4567$, which is the value of $z = x - y$ when represented in $\mathcal{F}(10, 5, -5, 5)$?

Scegli un'alternativa:

☐ a. 3.434 ☐ b. $0.3434 \cdot 10^0$ ☐ c. 3.4343

$$0.38917 \cdot 10^1$$

$$0.4567 \cdot 10^0$$

$$3,8917 -$$

$$0,4567 =$$

$$3,4350$$

Domanda 2

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Contrassegna
domanda

Given the matrix:

$$\begin{bmatrix} -1 & -1 \\ 0 & -2 \end{bmatrix} = \begin{bmatrix} -4 & 0 \\ -1 & -2 \end{bmatrix} = \begin{bmatrix} 17 & 2 \\ 2 & 4 \end{bmatrix}$$

Compute the 2-norm and the 1-norm of A :

Scegli un'alternativa:

☒ a. None of the above☐ b. $\|A\|_2 = 4, \|A\|_1 = \sqrt{5}$ ☐ c. $\|A\|_2 = 6, \|A\|_1 = 5$

$$(17 - \lambda)(4 - \lambda) - 4 = 0$$

$$(\lambda - 17)(\lambda - 4) - 4 = 0$$

$$\lambda^2 - 17\lambda - 4\lambda + 16 \cdot 4$$

$$\lambda^2 - 21\lambda + 64 = 0 \quad \lambda \notin \mathbb{N}$$

$$\|A\|_1 = \max\{|2|, |4|\} = 5$$

$$\|A\|_2 = \sqrt{\lambda_{\max}(A^T A)} =$$

Domanda 3

Risposta

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Contrassegna
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Which pairs of vectors are linearly independent?

Scegli un'alternativa:

☒ a. $(2, 1, 0), (-3, 4, 1)$ ☐ b. $(0, 1, -1), (0, \frac{1}{7}, -\frac{1}{7})$ ☐ c. $(\frac{1}{7}, \frac{1}{3}, 2), (1, \frac{2}{3}, 4)$

Domanda 4

Risposta

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Contrassegna
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If $x = (\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}})$, $y = (-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}})$, then:

Scegli un'alternativa:

☒ a. x and y are orthonormal☐ b. x and y are orthogonal☐ c. x and y are parallel

$$-\frac{1}{2} + \frac{1}{2} = 0$$

Domanda 5

Risposta

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Contrassegna
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If V is a vector space, $U \subseteq V$ is a subspace, and $\Pi_U : V \rightarrow U$ is a projection. Then:

Scegli un'alternativa:

☐ a. None of the above☒ b. $\Pi_U(x)$ is the point with minimum distance to x in U ☐ c. $\Pi_U(x)$ is the point with maximum distance to x in U

Domanda 6

Risposta

corretta

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If

$$A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \end{bmatrix}$$

Domanda 17

Risposta
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domandaIf X, Y are multivariate random variables, then:

Scegli un'alternativa:

- ☒ a. $Cov(X, Y) = -E[X]E[Y] + E[XY]$
- ☐ b. None of the above.
- ☐ c. $Cov(X, Y) = E[XY] - E[X]$

Domanda 18

Risposta
correttaPunteggio
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domandaGiven two random variables X and Y , Bayes Theorem implies that:

Scegli un'alternativa:

- ☐ a. $p(x) = \frac{p(y)p(y|x)}{p(y)}$
- ☐ b. $p(x) = \frac{p(x)p(y|x)}{p(y)}$
- ☒ c. $p(y) = \frac{p(y|x)p(x)}{p(x)}$

$$P(x|y) = \frac{P(y|x) \cdot P(x)}{P(y)}$$

$$P(x) = \frac{P(x|y) \cdot P(y)}{P(y|x)}$$

Domanda 19

Risposta
correttaPunteggio
ottenuto 1,00
su 1,00Contrassegna
domandaGiven two random variables X and Y such that $p(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$ and $p(y|x) = ce^{-|y-ax|}$, then the MLE reads:

Scegli un'alternativa:

- ☐ a. $x^* = \arg \min_x \frac{1}{2}(y - ax)^2$
- ☐ b. $x^* = \arg \min_x |y - ax| + x^2$
- ☒ c. $x^* = \arg \min_x |y - ax|$

Domanda 20

Risposta
correttaPunteggio
ottenuto 1,00
su 1,00Contrassegna
domandaThe regression matrix A of the data (x_i, y_i) , $i = 1, \dots, N$, $x_i, y_i \in \mathbb{R}$, for a polynomial model of degree $k - 1$, has shape:

Scegli un'alternativa:

- ☐ a. $k \times N$
- ☐ b. $N \times N$
- ☒ c. $N \times k$

Domanda 21

Risposta
correttaPunteggio
ottenuto 1,00
su 1,00Contrassegna
domandaIf $A \in \mathbb{R}^{n \times n}$ is symmetric and positive definite, then:

Scegli un'alternativa:

- ☐ a. All the eigenvalues λ_i of A are ≥ 0 .
- ☐ b. $A_{i,j} \geq 0$ for all i, j .
- ☒ c. None of the above.

Fine revisione

Domanda 11

Risposta
correttaPunteggio
ottenuto 1,00
su 1,00🚩
Contrassegna
domanda

If $f: \mathbb{R}^n \rightarrow \mathbb{R}$, $x^* \in \mathbb{R}^n$ is a strictly local minimum for f if:

Scegli un'alternativa:

- ☐ a. $\exists \epsilon > 0$ s.t. $f(x^*) \leq f(x), \forall x \in \mathbb{R}^n, \|x - x^*\| < \epsilon$.
- ☐ b. $f(x^*) \leq f(x), \forall x \in \mathbb{R}^n$.
- ☒ c. None of the above.

Domanda 12

Risposta
correttaPunteggio
ottenuto 1,00
su 1,00🚩
Contrassegna
domanda

If $f: \mathbb{R}^3 \rightarrow \mathbb{R}$, $f(x_1, x_2, x_3) = 2x_2 + x_1x_3 - 3x_1x_2$,
 $\nabla f(x_1, x_2, x_3) = (x_3 - 3x_2, 2 - 3x_1, x_1)$ then which of the following
statements is True?

Scegli un'alternativa:

- ☐ a. The gradient of f is equal to zero at multiple points.
- ☒ b. The gradient of f is never equal to zero.
- ☐ c. $(0, 0, 0)$ is a stationary point for f .

Domanda 13

Risposta errata

Punteggio
ottenuto 0,00
su 1,00🚩
Contrassegna
domanda

If $f: \mathbb{R}^n \rightarrow \mathbb{R}$, $f \in C^1(\mathbb{R}^n)$, then the Gradient Descent method
 $x_{k+1} = x_k - \alpha_k \nabla f(x_k)$ converges to a stationary point of f if:

Scegli un'alternativa:

- ☐ a. $\alpha_k > 0 \forall k \in \mathbb{N}$.
- ☒ b. $\alpha_k > 0$ is chosen with a backtracking procedure.
- ☐ c. $\alpha_k \rightarrow 0$ as $k \rightarrow \infty$.

Domanda 14

Risposta errata

Punteggio
ottenuto 0,00
su 1,00🚩
Contrassegna
domanda

Given two random variables X and Y , then:

Scegli un'alternativa:

- ☒ a. None of the above.
- ☐ b. $P(X = x|Y) \geq P(X = x)$.
- ☐ c. $P(X = x|Y) \leq P(X = x)$.

Domanda 15

Risposta
correttaPunteggio
ottenuto 1,00
su 1,00🚩
Contrassegna
domanda

The normal distribution $\mathcal{N}(\mu = \frac{1}{2}, \sigma^2 = 4)$ has the following PDF:

Scegli un'alternativa:

- ☐ a. $f_X(x) = \frac{1}{\sqrt{2\pi} \exp(-\frac{x^2}{2})}$.
- ☒ b. $f_X(x) = \frac{1}{2\sqrt{2\pi} \exp(-\frac{(x-\frac{1}{2})^2}{4})}$.
- ☐ c. $f_X(x) = \frac{1}{\sqrt{4\pi} \exp(-\frac{(x-\frac{1}{2})^2}{4})}$.

Domanda 16

Risposta
correttaPunteggio
ottenuto 1,00
su 1,00🚩
Contrassegna
domanda

Given two discrete random variable $X_1: \Omega \rightarrow \mathcal{T}$, $X_2: \Omega \rightarrow \mathcal{T}$ with
 $\mathcal{T} = \{0, 1\}$, and $f_{X_1} = \{\frac{1}{3}, \frac{2}{3}\}$, $f_{X_2} = \{\frac{2}{3}, \frac{1}{3}\}$ their PMF, then:

Scegli un'alternativa:

- ☐ a. $\mathbb{E}[X_1] = \mathbb{E}[X_2]$.
- ☐ b. $\mathbb{E}[X_1] < \mathbb{E}[X_2]$.
- ☐ c. $\mathbb{E}[X_1] > \mathbb{E}[X_2]$.

Domanda 6

Risposta
corretta
Punteggio
ottenuto 1,00
su 1,00
🚩
Contrassegna
domanda

If

$$A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

then:

Scegli un'alternativa:

- ☐ a. $\text{rank}(A) = 3$.
- ☐ b. $\text{rank}(A) = 2$.
- ☒ c. $\text{rank}(A) = 4$.

Domanda 7

Risposta
corretta
Punteggio
ottenuto 1,00
su 1,00
🚩
Contrassegna
domanda

If $A = U\Sigma V^T$ is the SVD decomposition of an $m \times n$ matrix A , then:

Scegli un'alternativa:

- ☐ a. None of the above.
- ☐ b. The elements on the diagonal of Σ are strictly positive.
- ☒ c. The elements of the diagonal matrix Σ are non-negative.

Domanda 8

Risposta errata
Punteggio
ottenuto 0,00
su 1,00
🚩
Contrassegna
domanda

The solution of $\min_x \|Ax - b\|_2^2$ where A is an $m \times n$ matrix, $m \geq n$, $\text{rank}(A) = n$, can be computed as:

Scegli un'alternativa:

- ☐ a. $A^\dagger x = b$.
- ☒ b. $x = A^+ b$.
- ☐ c. $AA^T x = A^T b$.

Domanda 9

Risposta
corretta
Punteggio
ottenuto 1,00
su 1,00
🚩
Contrassegna
domanda

If $f: \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = x_1 x_2$, $g: \mathbb{R} \rightarrow \mathbb{R}^2$, $g(t) = (t, t^2)$, then, if $h(t) = f(g(t))$:

Scegli un'alternativa:

- ☐ a. $h'(t) = 3t^3$.
- ☐ b. $h'(t) = t^2$.
- ☒ c. $h'(t) = 3t^2$.

Domanda 10

Risposta
corretta
Punteggio
ottenuto 1,00
su 1,00
🚩
Contrassegna
domanda

If $f: \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = x_1 + 2x_2^2$, $g: \mathbb{R} \rightarrow \mathbb{R}^2$, $g(t) = (t, t^2)$ then, if $h(t) = f(g(t))$,

Scegli un'alternativa:

- ☐ a. $h'(t) = t^3 + t$.
- ☐ b. $h'(t) = 8t^3 + 1$.
- ☒ c. $h'(t) = 3t^3 + 1$.

Domanda 11

If $f: \mathbb{R}^n \rightarrow \mathbb{R}$, $x^* \in \mathbb{R}^n$ is a strictly local minimum for f if:

Domanda 1

Risposta corretta

Punteggio
ottenuto 1,00 su
1,00Contrassegna
domanda

If $x = 0.7 \cdot 10^{-4}$ and $y = 1.7892$, which is the value of $z = x + y$ when represented in $\mathcal{F}(10, 5, -5, 5)$?

Scegli un'alternativa:

- ☒ a. None of the above.
- ☐ b. $0.17899 \cdot 10^1$.
- ☐ c. 0.17899.

Domanda 2

Risposta corretta

Punteggio
ottenuto 1,00 su
1,00Contrassegna
domanda

If A is symmetric and positive definite and $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$ are the eigenvalues of A , then:

Scegli un'alternativa:

- ☐ a. $\|A\|_2 = \sqrt{\lambda_1}$.
- ☒ b. $\|A\|_2 \lambda_1$.
- ☐ c. $\|A^{-1}\|_2 = \lambda_n$.

Domanda 3

Risposta corretta

Punteggio
ottenuto 1,00 su
1,00Contrassegna
domanda

The matrix:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$$



is:

Scegli un'alternativa:

- ☐ a. Semi-negative definite.
- ☐ b. Positive definite.
- ☒ c. Semi-positive definite.

$$(1-\lambda)(6-\lambda) - 6 = 0$$

$$(\lambda-1)(\lambda-6) - 6 = 0$$

$$\lambda^2 - 7\lambda + 6 = 0$$

$$\lambda(\lambda-7) = 0 \quad \begin{matrix} \lambda=0 \\ \lambda=7 \end{matrix}$$

Domanda 4

Risposta corretta

Punteggio
ottenuto 1,00 su
1,00Contrassegna
domanda

If $x = \begin{pmatrix} 1 \\ \sqrt{2} \end{pmatrix}$, $y = \begin{pmatrix} -1 \\ \sqrt{2} \end{pmatrix}$, then:

Scegli un'alternativa:

- ☐ a. None of the above.
- ☐ b. x and y are orthonormal.
- ☐ c. x and y are orthogonal.

Domanda 5

Risposta corretta

Punteggio
ottenuto 1,00 su
1,00Contrassegna
domanda

If V is a vector space with $\dim(V) = n$, $U \subseteq V$ is a subspace with $\dim(U) = k$, then:

Scegli un'alternativa:

- ☐ a. $U \cup U^\perp = U$.
- ☒ b. $U \cap U^\perp = \{0\}$.
- ☒ c. $U \cap U^\perp = \emptyset$.

Domanda 11

Risposta corretta

Punteggio
ottenuto 1,00 su
1,00Contrassegna
domandaIf $f : \mathbb{R}^n \rightarrow \mathbb{R}$. Which of the following statements is False?

Scegli un'alternativa:

- ☒ a. x^* local minimum for $f \implies \nabla f(x^*) = 0$.
- ☒ b. $\nabla f(x^*) = 0 \implies x^*$ local minimum for f
- ☒ c. $\nabla f(x^*) = 0 \implies x^*$ stationary point for f .

Domanda 12

Risposta corretta

Punteggio
ottenuto 1,00 su
1,00Contrassegna
domanda

If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = 2 \sin x_1 + \sin x_1 \cos x_2$,
 $\nabla f(x_1, x_2) = (2 \cos x_1 + \cos x_1 \cos x_2, -\sin x_1 \sin x_2)$ then which
 of the following is a stationary point for f ?

Scegli un'alternativa:

- ☒ a. $(-\frac{\pi}{2}, 0)$.
- ☒ b. None of the above.
- ☒ c. $(0, 0)$.

Domanda 13

Risposta corretta

Punteggio
ottenuto 1,00 su
1,00Contrassegna
domanda

If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = x_1^2 + \cos(x_2)$, then if the initial guess for a
 gradient descent iteration is $x^{(0)} = (1, \frac{\pi}{2})^T$ and $\alpha > 0$, then:

Scegli un'alternativa:

- ☒ a. $x^{(1)} = (1 + 2\alpha, \frac{\pi}{2} + \frac{\pi}{2}\alpha)^T$.
- ☒ b. $x^{(1)} = (1 - 2\alpha, \frac{\pi}{2} + \alpha)^T$.
- ☒ c. $x^{(1)} = (1 - 2\alpha, \frac{\pi}{2} + \frac{\pi}{2}\alpha)^T$.

Domanda 14

Risposta corretta

Punteggio
ottenuto 1,00 su
1,00Contrassegna
domandaGiven two random variables X and Y , then:

Scegli un'alternativa:

- ☒ a. $P(X = x|Y) \leq P(X = x)$.
- ☒ b. $P(X = x|Y) \geq P(X = x)$.
- ☒ c. None of the above.

Domanda 15

Risposta corretta

Punteggio
ottenuto 1,00 su
1,00Contrassegna
domandaFor a random variable $X : \Omega \rightarrow \mathcal{T}$, it holds:

Scegli un'alternativa:

- ☒ a. $Var(X) = E[X^2] - E[X]^2$.
- ☒ b. $Var(X) = E[X]^2 - E[X^2]$.
- ☒ c. $Var(X) = E[X^2] + E[X]^2$.

Domanda 16

Risposta corretta

Punteggio
ottenuto 1,00 su
1,00Contrassegna
domanda

Given two discrete random variable $X_1 : \Omega \rightarrow \mathcal{T}$, $X_2 : \Omega \rightarrow \mathcal{T}$ with
 $\mathcal{T} = \{0, 1\}$, and $f_{X_1} = \{\frac{1}{3}, \frac{2}{3}\}$, $f_{X_2} = \{\frac{2}{3}, \frac{1}{3}\}$ their PMF, then:

Scegli un'alternativa:

Domanda 16

Risposta corretta

Punteggio
ottenuto 1,00 su
1,00Contrassegna
domanda

Given two discrete random variable $X_1 : \Omega \rightarrow \mathcal{T}$, $X_2 : \Omega \rightarrow \mathcal{T}$ with $\mathcal{T} = \{0, 1\}$, and $f_{X_1} = \{\frac{1}{3}, \frac{2}{3}\}$, $f_{X_2} = \{\frac{2}{3}, \frac{1}{3}\}$ their PMF, then:

Scegli un'alternativa:

- ☒ a. $\mathbb{E}[X_1] > \mathbb{E}[X_2]$.
- ☐ b. $\mathbb{E}[X_1] = \mathbb{E}[X_2]$.
- ☐ c. $\mathbb{E}[X_1] < \mathbb{E}[X_2]$.

Domanda 17

Risposta corretta

Punteggio
ottenuto 1,00 su
1,00Contrassegna
domanda

If X, Y are multivariate random variables, then:

Scegli un'alternativa:

- ☐ a. None of the above.
- ☐ b. $Cov(X, Y) = -\mathbb{E}[X]\mathbb{E}[Y] + \mathbb{E}[X, Y]$.
- ☐ c. $Cov(X, Y) = \mathbb{E}[X, Y] - \mathbb{E}[X]$.

Domanda 18

Risposta corretta

Punteggio
ottenuto 1,00 su
1,00Contrassegna
domanda

Given two random variables X and Y , Bayes Theorem implies that

$$p(y|x) = \frac{p(x|y)p(y)}{p(x)} \text{ where:}$$

Scegli un'alternativa:

- ☐ a. $p(x|y)$ is called posterior distribution on y .
- ☐ b. $p(x|y)$ is called likelihood on y .
- ☐ c. $p(x|y)$ is called prior distribution on x .

Domanda 19

Risposta corretta

Punteggio
ottenuto 1,00 su
1,00Contrassegna
domanda

Given two random variables X and Y such that $p(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$ and $p(y|x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(y-ax)^2}$, then the MLE reads:

Scegli un'alternativa:

- ☐ a. $x^* = \arg \min_x \frac{1}{2}(y-ax)^2 + x^2$.
- ☐ b. $x^* = \arg \min_x \frac{1}{2}(y-ax)^2 + \frac{1}{2}x^2$.
- ☐ c. $x^* = \arg \min_x \frac{1}{2}(y-ax)^2$.

Domanda 20

Risposta corretta

Punteggio
ottenuto 1,00 su
1,00Contrassegna
domanda

Suppose a set of data (x_i, y_i) , $i = 1, \dots, N$, $y_i = f(x_i) + \epsilon_i$, where $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$. In linear regression, the likelihood function is:

Scegli un'alternativa:

- ☐ a. $p(y|x, \theta) = \prod_{i=1}^N \mathcal{N}(y|\theta^T x_i, \sigma^2)$.
- ☐ b. $p(y|x, \theta) = \prod_{i=1}^N \mathcal{N}(y_i|\theta^T x_i, \sigma^2)$.
- ☐ c. $p(y|x, \theta) = \prod_{i=1}^N \mathcal{N}(x|\theta^T x_i, \sigma^2)$.

Domanda 21

Risposta corretta

If $f : \mathbb{R}^3 \rightarrow \mathbb{R}$, $f(x_1, x_2, x_3) = 2x_2 + x_1x_3 - 3x_1x_2$.

Domanda 21

Risposta corretta

Punteggio
ottenuto 1,00 su
1,00🚩 Contrassegna
domanda

If $f : \mathbb{R}^3 \rightarrow \mathbb{R}$, $f(x_1, x_2, x_3) = 2x_2 + x_1x_3 - 3x_1x_2$,
 $\nabla f(x_1, x_2, x_3) = (x_3 - 3x_2, 2 - 3x_1, x_1)$ then which of the following
is a stationary point for f ?

Scegli un'alternativa:

- ☒ a. $(0, 0, 0)$. ☐
- ☒ b. None of the above. ☐
- ☒ c. $(0, 2, 0)$. ☐



Domanda 1

Risposta
correttaPunteggio
ottenuto 1,00
su 1,00P
Contrassegna
domanda

If $x = 3.89167$ and $y = 0.4567$, which is the value of $z = x - y$ when represented in $F(10, 5, -5, 5)$?

Scegli un'alternativa:

- ☐ a. 3.4343.
- ☐ b. $0.3434 \cdot 10^0$.
- ☐ c. 3.434.

Domanda 2

Risposta
correttaPunteggio
ottenuto 1,00
su 1,00P
Contrassegna
domanda

A is positive definite if:

Scegli un'alternativa:

- ☒ a. All its eigenvalues are > 0 .
- ☐ b. All its eigenvalues are ≥ 0 .
- ☐ c. None of the above.

Domanda 3

Risposta
correttaPunteggio
ottenuto 1,00
su 1,00P
Contrassegna
domanda

The matrix:

$$A = \begin{bmatrix} 3 & 4 \\ 4 & 9 \end{bmatrix}$$

is:

Scegli un'alternativa:

- ☐ a. None of the above.
- ☒ b. Symmetric and positive definite.
- ☐ c. Symmetric and semi-positive definite.

$$(3-\lambda)(9-\lambda) - 16 = 0$$

$$(3-\lambda)(\lambda-9) - 16 = 0$$

$$\lambda^2 - 12\lambda + 8 = 0$$

$$\lambda = 6 \pm \sqrt{36 - 8} = 6 \pm 5 = 11$$

Domanda 4

Risposta
correttaPunteggio
ottenuto 1,00
su 1,00P
Contrassegna
domanda

If U is an $n \times n$ orthogonal matrix, $x \in \mathbb{R}^n$, then:

Scegli un'alternativa:

- ☐ a. $U^T = U$.
- ☒ b. $\|Ux\| = \|x\|$.
- ☐ c. $U^2 = U$.

Domanda 5

Risposta
correttaPunteggio
ottenuto 1,00
su 1,00P
Contrassegna
domanda

If V is a vector space with $\dim(V) = n$, $U \subseteq V$ is a subspace with $\dim(U) = k$, then:

Scegli un'alternativa:

- ☐ a. $\dim(U^\perp) = k$.
- ☐ b. None of the above.
- ☒ c. $\dim(U^\perp) = n - k$.

Domanda 1

Risposta errata

Punteggio
ottenuto 0,00
su 1,00Contrassegna
domandaThe real number $x = 79.54$ in normalized scientific representation is:

Scegli un'alternativa:

- ☐ a. $fl(x) = 0.7954 \cdot 10^2$
- ☐ b. None of the above.
- ☐ c. $fl(x) = 0.7954 \cdot 10^3$

Domanda 2

Risposta
correttaPunteggio
ottenuto 1,00
su 1,00Contrassegna
domanda

Given the matrix:

$$\| \cdot \|_1 = 5$$

$$\begin{bmatrix} -1 & -1 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} -4 & 0 \\ -1 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

Compute the 2-norm and the 1-norm of A : $(\lambda - 1)(\lambda - 1) - 4 = 0$

Scegli un'alternativa:

$$\lambda^2 - 2\lambda + 6 = 0$$

- ☐ a. $\|A\|_2 = 6, \|A\|_1 = \sqrt{7}$
- ☒ b. None of the above.
- ☐ c. $\|A\|_2 = 4, \|A\|_1 = 5$

Domanda 3

Risposta
correttaPunteggio
ottenuto 1,00
su 1,00Contrassegna
domanda

The matrix:

$$A = \begin{bmatrix} 3 & 4 \\ 4 & 9 \end{bmatrix}$$

is:

Scegli un'alternativa:

- ☐ a. Symmetric and positive definite.
- ☐ b. Symmetric and semi-positive definite.
- ☐ c. None of the above.

Domanda 4

Risposta
correttaPunteggio
ottenuto 1,00
su 1,00Contrassegna
domandaIf $x = (3, 1)$, $y = (-3, -1)$, then:

Scegli un'alternativa:

- ☐ a. x and y are orthogonal.
- ☐ b. x and y are orthonormal.
- ☐ c. None of the above.

Domanda 5

Risposta
correttaPunteggio
ottenuto 1,00
su 1,00Contrassegna
domandaIf V is a vector space with $\dim(V) = n$, $U \subseteq V$ is a subspace with $\dim(U) = k$, then:

Scegli un'alternativa:

- ☐ a. $U \cap U^\perp = \emptyset$.
- ☐ b. $U \cap U^\perp = \{0\}$.
- ☐ c. $U \cup U^\perp = U$.

Domanda 6

Risposta

corretta

Punteggio

ottenuto 1,00

su 1,00

🚩

Contrassegna domanda

If

$$A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

then:

Scegli un'alternativa:

- ☐ a. $\text{rank}(A) = 4$.
- ☐ b. $\text{rank}(A) = 2$.
- ☐ c. $\text{rank}(A) = 3$.

Domanda 7

Risposta

corretta

Punteggio

ottenuto 1,00

su 1,00

🚩

Contrassegna domanda

If $A = U\Sigma V^T$ is the SVD decomposition of an $m \times n$ matrix A , then:

Scegli un'alternativa:

- ☐ a. The singular values σ_i of A are $\sigma_i = \sqrt{\lambda_i(A^T A)}$ where $\lambda_i(A^T A)$ are the eigenvalues of $A^T A$.
- ☐ b. None of the above.
- ☒ c. The singular values σ_i of A are $\sigma_i = \lambda_i(A^T A)$ where $\lambda_i(A^T A)$ are the eigenvalues of $A^T A$.

Domanda 8

Risposta

corretta

Punteggio

ottenuto 1,00

su 1,00

🚩

Contrassegna domanda

The problem $\min_x \|Ax - b\|_2^2$ where A is an $m \times n$ matrix, $m \geq n$, $\text{rank}(A) = k$,

Scegli un'alternativa:

- ☐ a. Has a unique solution if and only if $k < n$.
- ☐ b. Has a unique solution if and only if $k = n$.
- ☐ c. None of the above.

Domanda 9

Risposta

corretta

Punteggio

ottenuto 1,00

su 1,00

🚩

Contrassegna domanda

If $f: \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x, y) = \sqrt{x^2 - y^2}$, $g: \mathbb{R} \rightarrow \mathbb{R}^2$, $g(t) = (e^{2t}, e^{-t})$ then, if $h(t) = f(g(t))$,

Scegli un'alternativa:

- ☐ a. $h'(t) = \frac{2e^{4t} - 1}{e^t \sqrt{e^{4t} - 1}}$.
- ☐ b. $h'(t) = \frac{2e^{4t} - e^{-2t}}{\sqrt{e^{4t} + e^{-2t}}}$.
- ☐ c. $h'(t) = \frac{2e^{4t} + e^{-2t}}{\sqrt{e^{4t} - e^{-2t}}}$.

Domanda 10

Risposta

corretta

Punteggio

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Contrassegna domanda

If $f: \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = x_1 + 2x_2^2$, $g: \mathbb{R} \rightarrow \mathbb{R}^2$, $g(t) = (t, t^2)$ then, if $h(t) = f(g(t))$,

Scegli un'alternativa:

- ☐ a. $h'(t) = t^3 + t$.
- ☐ b. $h'(t) = 3t^3 + 1$.
- ☐ c. $h'(t) = 8t^3 + 1$.



Domanda 11

Risposta

corretta

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su 1,00Contrassegna
domanda

If $f: \mathbb{R}^n \rightarrow \mathbb{R}$, $x^* \in \mathbb{R}^n$ is a stationary point for f :

Scegli un'alternativa:

- ☒ a. None of the above.
- ☐ b. if $\nabla f(x^*) < 0$.
- ☐ c. if $\nabla f(x^*) = 0$.

Domanda 12

Risposta

corretta

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domanda

Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = 9x_1x_2^2 - x_1$,
 $\nabla f(x_1, x_2) = (9x_2^2 - 1, 18x_1x_2)$ then which of the following
 statements is True?

Scegli un'alternativa:

- ☒ a. f has 2 stationary points.
- ☐ b. f has 1 stationary point.
- ☐ c. f has 0 stationary points.

Domanda 13

Risposta

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domanda

If $f: \mathbb{R}^n \rightarrow \mathbb{R}$, $f \in C^1(\mathbb{R}^n)$, then x^* is said to be a stationary point of f if:

Scegli un'alternativa:

- ☒ a. The Hessian of f in x^* is positive definite.
- ☐ b. $\nabla f(x^*) = 0$.
- ☐ c. x^* is a minimum point of f .

Domanda 14

Risposta

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domanda

If Ω is the sample space, $\mathcal{A}$ is the event space and $\mathcal{T}$ is a subset of $\mathbb{R}$, a
 random variable X is:

Scegli un'alternativa:

- ☒ a. A function $X: \Omega \rightarrow \mathcal{T}$.
- ☐ b. A function $X: \mathcal{A} \rightarrow \mathcal{T}$.
- ☐ c. A function $X: \Omega \rightarrow \mathcal{A}$.

Domanda 15

Risposta

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domanda

If $X: \Omega \rightarrow \mathcal{T}$ is a discrete random variable with PMF $f_X: \mathcal{T} \rightarrow [0, 1]$,
 then:

Scegli un'alternativa:

- ☒ a. $\mathbb{E}[X] = \sum_{i \in \mathcal{T}} i$.
- ☐ b. $\mathbb{E}[X] = \sum_{i \in \mathcal{T}} i f_X(i)$.
- ☐ c. $\mathbb{E}[X] = \sum_{i \in \mathcal{T}} f_X(i)$.

Domanda 16

Risposta

Given a discrete random variable $X: \Omega \rightarrow \mathcal{T}$, with $\mathcal{T} = \{1, 2, 3\}$, and
 $f_X = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ its PMF, then:

Domanda 16

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Contrassegna
domanda

Given a discrete random variable $X: \Omega \rightarrow \mathcal{T}$, with $\mathcal{T} = \{1, 2, 3\}$, and $f_X = \{\frac{1}{6}, \frac{1}{3}, \frac{1}{2}\}$ its PMF, then:

Scegli un'alternativa:

- ☐ a. $\mathbb{E}[X] = 6$.
- ☐ b. $\mathbb{E}[X] = 2$.
- ☒ c. $\mathbb{E}[X] = \frac{11}{6}$.

Domanda 17

Risposta errata

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Contrassegna
domanda

If X is a random variable with values in $\mathbb{R}^D$, V_x is the variance of X , then:

Scegli un'alternativa:

- ☐ a. $V_x[X]$ is a vector $D \times 1$.
- ☐ b. $V_x[X]$ is a matrix $D \times D$.
- ☒ c. $V_x[X]$ is a scalar.

Domanda 18

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domanda

The quantity of interest in Bayes' Theorem is:

Scegli un'alternativa:

- ☐ a. The posterior.
- ☐ b. The marginal.
- ☒ c. The likelihood.

Domanda 19

Risposta
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domanda

Given two random variables X and Y such that $p(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$ and $p(y|x) = ce^{-|y-ax|}$, then the MAP reads:

Scegli un'alternativa:

- ☒ a. $x^* = \arg \min_x |y - ax| + \frac{1}{2}x^2$.
- ☐ b. $x^* = \arg \min_x \frac{1}{2}(y - ax)^2$.
- ☐ c. $x^* = \arg \min_x |y - ax|$.

Domanda 20

Risposta errata

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Contrassegna
domanda

Suppose a set of data (x_i, y_i) , $i = 1, \dots, N$, $y_i = f(x_i) + \epsilon_i$, where $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$. In linear regression, the likelihood function is:

Scegli un'alternativa:

- ☐ a. None of the above.
- ☐ b. $p(y|x, \theta) = \prod_{i=1}^N \mathcal{N}(x|x_i^T \theta, \sigma^2)$.
- ☒ c. $p(y|x, \theta) = \prod_{i=1}^N \mathcal{N}(y|x_i^T \theta, \sigma^2)$.

Domanda 21

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domanda

If $f: \mathbb{R}^3 \rightarrow \mathbb{R}$, $f(x_1, x_2, x_3) = x_1^2 - 3x_2^2 x_1 - x_3 x_2$, then $\nabla f(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ equals to:

Scegli un'alternativa:

- ☐ a. $(\frac{11}{4}, -1, \frac{1}{2})$.
- ☐ b. $(\frac{11}{4}, 1, \frac{1}{2})$.
- ☒ c. $(\frac{11}{4}, \frac{1}{2}, -1)$.

Domanda 6

Risposta corretta

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1,00Contrassegna
domanda

If

$$A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

then:

Scegli un'alternativa:

- ☒ a. $\text{rank}(A) = 2$
- ☐ b. $\text{rank}(A) = 4$
- ☐ c. $\text{rank}(A) = 3$

Domanda 7

Risposta corretta

Punteggio
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domandaIf $A = U\Sigma V^T$ is the SVD decomposition of an $m \times n$ matrix A , then:

Scegli un'alternativa:

- ☐ a. The columns of U are eigenvectors of AA^T .
- ☐ b. The rows of V^T are eigenvectors of AA^T .
- ☐ c. None of the above

Domanda 8

Risposta corretta

Punteggio
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1,00Contrassegna
domandaThe problem $\min_x \|Ax - b\|_2^2$ where A is an $m \times n$ matrix, $m \geq n$, $\text{rank}(A) = k$,

Scegli un'alternativa:

- ☐ a. Has infinite solutions.
- ☐ b. Has a unique solution if $k = n$.
- ☐ c. Has a unique solution for any k .

Domanda 9

Risposta corretta

Punteggio
ottenuto 1,00 su
1,00Contrassegna
domandaIf $f: \mathbb{R}^3 \rightarrow \mathbb{R}$, $f(x_1, x_2, x_3) = \sin x_1 - \sin x_2 \cos x_3 + x_3^2$, then $\nabla f(\frac{\pi}{2}, \frac{\pi}{2}, \pi)$ equals to:

Scegli un'alternativa:

- ☐ a. $(1, 0, 2\pi)$.
- ☐ b. $(0, 0, \pi)$.
- ☐ c. $(0, 0, 2\pi)$.

Domanda 10

Risposta corretta

Punteggio
ottenuto 1,00 su
1,00Contrassegna
domandaIf $f: \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = x_1 x_2$, $g: \mathbb{R} \rightarrow \mathbb{R}^2$, $g(t) = (t, t^2)$ then, if $h(t) = f(g(t))$,

Scegli un'alternativa:

- ☐ a. $h'(t) = t^2 + 1$.
- ☐ b. $h'(t) = 2t^2$.
- ☒ c. $h'(t) = 3t^2$.